THE UNIVERSAL FIRST ORDER THICKENING OF A FORMALLY UNRAMIFIED MORPHISM IS FORMALLY UNRAMIFIED OVER THE TARGET

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3184-3190 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 3192-3215, the complete author proof (22 body lines).

Editorial note. Navigation stubs: 0 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Formally unramified morphisms of algebraic spaces, universal first order thickenings, and scheme-theoretic inverse images; the two applications are retained.

Lemma 0.1. *Let S be a scheme. Let $Z \to X$ be a formally unramified morphism of algebraic spaces over S. Then the universal first order thickening Z' is formally unramified over X.*

Proof. Let $T \subset T'$ be a first order thickening of affine schemes over X. Let

$$\begin{array}{ccc} Z' & \xleftarrow{c} & T \\ \downarrow & \nwarrow^{a,b} & \downarrow \\ X & \longleftarrow & T' \end{array}$$

be a commutative diagram. Set $T_0 = c^{-1}(Z) \subset T$ and $T'_a = a^{-1}(Z)$ (scheme theoretically). Since Z' is a first order thickening of Z, we see that T' is a first order thickening of T'_a. Moreover, since $c = a|_T$ we see that $T_0 = T \cap T'_a$ (scheme theoretically). As T' is a first order thickening of T it follows that T'_a is a first order thickening of T_0. Now $a|_{T'_a}$ and $b|_{T'_a}$ are morphisms of T'_a into Z' over X which agree on T_0 as morphisms into Z. Hence by the universal property of Z' we conclude that $a|_{T'_a} = b|_{T'_a}$. Thus a and b are morphism from the first order thickening T' of T'_a whose restrictions to T'_a agree as morphisms into Z. Thus using the universal property of Z' once more we conclude that $a = b$. In other words, the defining property of a formally unramified morphism holds for $Z' \to X$ as desired. $\square$

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.